



Dow Packaging & Speciality Plastics Product Data Sheet

APPEEL™ 55010

Peelable Resin

Description			
Product Description	APPEEL™ 55010 is a modified polyolefin resin designed to function as a sealing layer for lidding applications, most commonly sealing to polypropylene. It is available in pellet form for use in conventional extrusion or coextrusion equipment designed to process polyethylene resins.		
Restrictions			
Material Status	Commercial: Active		
Typical Characteristics			
Uses	Lidding Sealant		
Applications	- APPEEL™ 55010 has strong sealability to PP and allows a peelable seal from it. - Boiling resistance. APPEEL™ 55010 endures boiling process. - In the case of: OPET/ adh // APPEEL™ 55010 , these film structures endure boiling condition at 120C / 30min - APPEEL™ 55010 conforms to Code #20 of the Ministry of Health and Welfare Japan. Typical structures for this lidding would be: OPET / adhesive // APPEEL™ 55010 OPET / adhesive // coex film with APPEEL™ 55010 APPEEL™ 55010 is used as a heat seal layer in lidding material for injection molded and vacuum molded PP containers used in the packaging of puddings, jelly, and other food items.		
Typical Properties			
Physical	Nominal Values		Test Method(s)
*Density ()	0.92 g/cm³	ASTM D792	ISO 1183
*Melt Flow Rate (190°C/2.16kg)	2 g/10 min	ASTM D1238	ISO 1133
Thermal	Nominal Values		Test Method(s)
*Melting Point (DSC)	143 °C (289.4 °F)	ASTM D3418	ISO 3146
Vicat Softening Point ()	99 °C (210.2 °F)	ASTM D1525	ISO 306
Heat Seal Evaluation	The performance of any sealant resin should be evaluated within the context of the application. The sealant is designed to bond to particular substrate(s). Many variables can affect seal strength, including the physical properties of the substrate being sealed to, thickness, flange or surface design, heat seal temperature, dwell time and pressure. The condition and type of the sealing equipment used, such as roller sealers versus platen seal mechanisms can make a significant difference. In most cases sealant peel strength is used as a measure of performance. Although this is a convenient test, peel strength is affected not only by substrate adhesion but also by peel angle, separation rate, ambient temperature, tensile and modulus properties of the materials, and often by the time elapsed since the formation of the bond. If sealant peel strength is used as a measure of sealant performance, it is imperative that peel strength be evaluated not only at the time of initial heat sealing the lid to the substrate, but throughout the life of the product and under all the conditions to which the sealant will be exposed. Only then does peel strength provide a reliable indication of adhesive performance in the specific application.		
Processing Information			
*Maximum Processing Temperature	285 °C (545 °F)		
General Processing Information	If the process is stopped for short periods of time, the screw for the APPEEL™ extruder should be kept turning at a low rpm to keep material flowing.		

After processing APPEEL™ purge the material out using a polyethylene resin, preferably with a lower melt flow rate than the APPEEL™ resin in use. The "Disco Purge Method" is suggested as the preferred purging method, as this method usually results in a more effective purging process. Information on the Disco Purge Method can be obtained via your Dow Sales Representative.

Never shut down the extrusion system with APPEEL™ in the extruder and die. Properly purge out the APPEEL™ with a polyethylene, and shut down the line with polyethylene or polypropylene in the system.

Blown Film Processing Processing Information

Nominal Values

Blown Film: The melt temperature of APPEEL™ 55010 should be maintained in the 160 - 185°C range. Selection of a specific melt temperature will depend on screw configuration, potential power limitations, and the need to match melt viscosities. However, melt temperatures above 285°C should be avoided because of possible thermal degradation of the resin.

If the process is stopped for short periods of time, the APPEEL™ 55010 resin extruder should be kept turning at low rpm. For a permanent shutdown, the APPEEL™ 55010 resin should be purged out using an available polyethylene resin run at the same extrusion temperature used for the APPEEL™ 55010 resin. Never raise temperature over 285°C until APPEEL™ 55010 resin is completely purged out. APPEEL™ 55010 requires relatively low processing temperatures and cooling the bottom of hopper due to its low Vicat point and higher comonomer level.

Following is an example for suggested temperature profile on the low side of the processing range. Higher temperatures in the final metering zone, adapter and die are suggested if compatible with the process and application.

Feed Zone	135 °C (275 °F)
Second Zone	160 °C (320 °F)
Third Zone	160 °C (320 °F)
Fourth Zone	160 °C (320 °F)
Fifth Zone	160 °C (320 °F)
Adapter Zone	160 °C (320 °F)
Die Zone	160 °C (320 °F)

Cast Film / Sheet Processing Processing Information

Nominal Values

Cast Film: The melt temperature of APPEEL™ 55010 should be maintained in the 210 - 235°C range. Selection of a specific melt temperature will depend on screw configuration, potential power limitations, and the need to match melt viscosities. However, melt temperatures above 285°C should be avoided because of possible thermal degradation of the resin.

If the process is stopped for short periods of time, the APPEEL™ 55010 resin extruder should be kept turning at low rpm. For a permanent shutdown, the APPEEL™ 55010 resin should be purged out using an available polyethylene resin run at the same extrusion temperature used for the APPEEL™ 55010 resin. Never raise temperature over 285°C until APPEEL™ 55010 resin is completely purged out. APPEEL™ 55010 requires relatively low processing temperatures and cooling the bottom of hopper due to its low Vicat point and higher comonomer level.

Following is an example for suggested temperature profile on the high side of the processing range. Lower temperatures in the final metering zone, adapter and die are suggested if compatible with the process and application.

Feed Zone	160 °C (320 °F)
Second Zone	185 °C (365 °F)
Third Zone	210 °C (410 °F)
Fourth Zone	235 °C (455 °F)
Fifth Zone	235 °C (455 °F)
Adapter Zone	235 °C (455 °F)
Die Zone	235 °C (455 °F)

FDA Status Information

APPEEL™ 55010 Lidding Sealant Resin complies with Food and Drug Administration Regulation 21 CFR 177.1520(c)3.2a subject to extractive limitations of 177.1330(c)

The information and certifications provided herein are based on data we believe to be reliable, to the best of our knowledge. The information and certifications apply only to the specific material designated herein as sold by Dow and do not apply to

use in any process or in combination with any other material. They are provided at the request of and without charge to our customers. Accordingly, Dow cannot guarantee or warrant such certifications or information and assumes no liability for their use.

Regulatory Information

APPEEL™ 55010 complies with Japan Hygienic Olefin and Styrene Plastics Association and MITI no. 20 Food regulation in Japan.

For information on regulatory compliance outside of the U.S.A. and Japan, consult your local Dow representative.

Safety & Handling

For information on appropriate Handling & Storage of this polymeric resin, please refer to the material Safety Data Sheet.

A Product Safety Bulletin, Material Safety Data Sheet, and/or more detailed information on extrusion processing and/or compounding of this polymeric resin for specific applications are available from your Dow representative.

Product Stewardship

The Dow Chemical Company and its subsidiaries (“Dow”) has a fundamental concern for all who make, distribute, and use its products, and for the environment in which we live. This concern is the basis for our Product Stewardship philosophy by which we assess the safety, health, and environmental information on our products and then take appropriate steps to protect employee and public health and our environment. The success of our Product Stewardship program rests with each and every individual involved with Dow products — from the initial concept and research, to manufacture, use, sale, disposal, and recycle of each product.

Customer Notice

Dow strongly encourages its customers to review both their manufacturing processes and their applications of Dow products from the standpoint of human health and environmental quality to ensure that Dow products are not used in ways for which they are not intended or tested. Dow personnel are available to answer your questions and to provide reasonable technical support. Dow product literature, including safety data sheets, should be consulted prior to use of Dow products. Current safety data sheets are available from Dow.

Medical Applications Policy

NOTICE REGARDING MEDICAL APPLICATION RESTRICTIONS: Dow will not knowingly sell or sample any product or service (“Product”) into any commercial or developmental application that is intended for:

- a. long-term or permanent contact with internal bodily fluids or tissues. “Long-term” is contact which exceeds 72 continuous hours (or for PELLETHANE™ Polyurethane Elastomers only, which exceeds 30 days);
- b. use in cardiac prosthetic devices regardless of the length of time involved (“cardiac prosthetic devices” include, but are not limited to, pacemaker leads and devices, artificial hearts, heart valves, intra-aortic balloons and control systems, and ventricular bypass-assisted devices);
- c. use as a critical component in medical devices that support or sustain human life; or
- d. use specifically by pregnant women or in applications designed specifically to promote or interfere with human reproduction.

Dow requests that customers considering use of Dow products in medical applications notify Dow so that appropriate assessments may be conducted.

Dow does not endorse or claim suitability of its products for specific medical applications. It is the responsibility of the medical device or pharmaceutical manufacturer to determine that the Dow product is safe, lawful, and technically suitable for the intended use. **DOW MAKES NO WARRANTIES, EXPRESS OR IMPLIED, CONCERNING THE SUITABILITY OF ANY DOW PRODUCT FOR USE IN MEDICAL APPLICATIONS.**

Disclaimer

NOTICE: No freedom from infringement of any patent owned by Dow or others is to be inferred. Because use conditions and applicable laws may differ from one location to another and may change with time, the Customer is responsible for determining whether products and the information in this document are appropriate for the Customer’s use and for ensuring that the Customer’s workplace and disposal practices are in compliance with applicable laws and other governmental enactments. Dow assumes no obligation or liability for the information in this document. **NO WARRANTIES ARE GIVEN; ALL IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE ARE EXPRESSLY EXCLUDED.**

NOTICE: If products are described as “experimental” or “developmental”: (1) product specifications may not be fully determined; (2) analysis of hazards and caution in handling and use are required; (3) there is greater potential for Dow to change specifications and/or discontinue production; and (4) although Dow may from time to time provide samples of such products, Dow is not obligated to supply or otherwise commercialize such products for any use or application whatsoever.

Additional Information

North America

U.S. & Canada: **1-800-441-4369**
1-989-832-1426
Mexico: **+1-800-441-4369**

South Africa **+800-99-5078**

Europe/Middle East

All Countries **+31-11567-2626**
+800-3694-6367
Italy: **+800-783-825**

Asia Pacific **+800-7776-7776**
+60-3-7958-5392

Latin America

Argentina: **+54-11-4319-0100**
Brazil: **+55-11-5188-9000**
Colombia: **+57-1-219-6000**
Mexico: **+52-55-5201-4700**

<http://www.dow.com>

Published August 2019

© 2019 The Dow Chemical Company

Version: 230.0

Last modified at 8/6/2019 6:04 AM